



Aim:- To develop a basic dynamic web application using Servlet technology.

Objectives:-

- To understand Servlet lifecycle
- To handle HTTP requests and responses.
- To generate dynamic web content.

Concepts Used:-

- Servlet (HttpServlet)
- Methods: doGet(), doPost()
- HTTP Protocol
- Apache Tomcat Server

Theory

A servlet is a server-side java program that processes client requests and generate dynamic responses. When a client sends a request:

1. Server receives it
2. Servlet processes it.
3. Response is generated and sent back.

Servlet lifecycle:

init():- initialization

service():- request handling

destroy():- cleanup

Output

Hello, this is Servlet Web App.



Algorithm

1. Start Tomcat server.
2. Create Servlet class
3. Override doGet() method
4. Accepts client request
5. Generate response using PrintWriter
6. Send response to browser
7. Display Output.

Source Code :-

```

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class HelloServlet extends HttpServlet {
    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html><body>");
        out.println("<h2>Hello, this is Servlet. Web App</h2>");
        out.println("</body></html>");
    }
}

```




Result :-

The Servlet-based web application is successfully executed and displays dynamic output in the browser.

Completed
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